RAG Assignment: Baseline vs. Re-Ranking

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Code: `https://github.com/meet6868/RAG_Biencoder_CrossEncoder`

Setup

Document: the EU General Data Protection Regulation (GDPR), an 88-page legal PDF, split into 559 chunks. **LLM:** `Qwen2.5-3B-Instruct` (local). Both pipelines use the same chunks and the same prompt.

- **Pipeline 1 (baseline):** the Bi-Encoder `all-MiniLM-L6-v2` finds the Top-10 chunks by cosine similarity; the Top-3 go to the LLM → Answer 1.
- **Pipeline 2 (re-ranking):** the Cross-Encoder `ms-marco-MiniLM-L-6-v2` re-scores the same 10 chunks; the new Top-3 go to the LLM → Answer 2.

1 Architecture comparison

Bi-Encoder: the question and each chunk are turned into vectors *separately*, and we compare the vectors with cosine similarity. Chunk vectors are made once in advance, so search is very fast. But a whole chunk is squeezed into one vector, so small words like *not* or *unless* are easily lost.

Cross-Encoder: the question and one chunk go into the model *together*, so every word of the question can look at every word of the chunk (self-attention). It judges relevance much better, but it must run once for every (question, chunk) pair, so it is slow.

2 Qualitative and rank-shift analysis

For each question, the "gold" chunk is the chunk that contains the answer in the law. Only ranks 1–3 are given to the LLM.

ID	Failure type	Question	Gold chunk	Rank shift
Q1	Keyword trap	What is the highest penalty for a company that violates people's rights under GDPR?	Art. 83(5)	#4 → #1
Q2	Exception	Do I have to notify affected users of a breach if the stolen data was encrypted?	Art. 34(3)	#4 → #1
Q3	Negation / role limit	Can the data protection officer be fired for doing their job?	Art. 38(3)	#5 → #1
Q4	Buried at rank 4–10	Does a company have to stop processing if a person objects?	Art. 21(1)	#9 → #1
Q5	Keyword trap (synonym)	How quickly must a company fix wrong information about me?	Art. 16	#9 → #3

Example (Q1). The word "penalty" made the Bi-Encoder put Article 84 (titled "Penalties") first, but the real answer, Art. 83(5), was only 4th. The Cross-Encoder moved it to 1st:

Rank	Bi-Encoder	Cross-Encoder
1	Art. 84 "Penalties" (wrong)	**Art. 83(5) – gold**
2	Art. 83(4) (lower fine tier)	Art. 83(4)
3	Recital 150	Recital 150
4	**Art. 83(5) – gold**	Art. 84 "Penalties"

Answer 1 vs. Answer 2 (exact LLM answers for the questions where the Bi-Encoder failed):

Q1 – Keyword trap	
Question	What is the highest penalty for a company that violates people's rights under GDPR?
Answer 1 (Bi-Encoder)	The highest penalty for a company that violates people's rights under GDPR is 2% of the company's global annual turnover of the previous financial year, whichever is higher. [Art. 83(4)]
Answer 2 (Cross-Encoder)	The highest penalty for a company violating people's rights under GDPR is 4% of the company's global annual turnover, or up to 20,000,000 EUR, whichever is greater. [Art. 83(5)]
Result	Answer 1 **misstates the rule**: it gives the lower 2% tier from Art. 83(4). Answer 2 is correct: EUR 20 million or 4%, whichever is higher.
Why	The question says "penalty", and Art. 84 is titled "Penalties", so the Bi-Encoder ranked it 1st, even though it only lets countries add their own penalties. Art. 83(4), about a lower fine, also looked similar. The right chunk, Art. 83(5), uses different words ("administrative fines", "data subjects' rights"), so it was only 4th. The Cross-Encoder reads the question and the chunk together, so it matched "people's rights" with "data subjects' rights" and put Art. 83(5) 1st.

Q2 – Exception	
Question	Do I have to notify affected users of a breach if the stolen data was encrypted?
Answer 1 (Bi-Encoder)	The provided context does not directly state whether encryption of data affects the notification obligation. However, it mentions that if the controller can demonstrate that the personal data breach is unlikely to result in a risk to the rights and freedoms of natural persons, they are not required to notify the supervisory authority. [Art. 33(2)]
Answer 2 (Cross-Encoder)	Yes, according to Art. 34(3)(a), if the controller has implemented appropriate technical and organizational measures, such as encryption, and those measures were applied to the personal data affected by the breach, then notification to the data subjects is not required.
Result	Answer 1 **misses the exception**. Answer 2 is correct: with encryption, telling the users is not required.
Why	The Bi-Encoder's Top-3 were the general data-breach rules (Art. 33 and Recitals 85 and 87), which share many words with the question ("notify", "breach"). The exception (Art. 34(3)) is about the same topic, and the deciding word "encryption" is only one word in a long chunk, so it hardly changed the chunk's vector. It ended up 4th. The Cross-Encoder saw "encrypted" in the question next to "such as encryption" in the chunk and ranked the exception 1st.

Q3 – Negation / role limit	
Question	Can the data protection officer be fired for doing their job?
Answer 1 (Bi-Encoder)	According to Art. 38(5), the data protection officer is bound by secrecy concerning the performance of their tasks, in accordance with Union or Member State law. This implies that the officer cannot be fired for doing their job as it would involve breaching confidentiality. [Art. 38(5)]
Answer 2 (Cross-Encoder)	Yes, the data protection officer cannot be dismissed or penalized by the controller or processor for performing their tasks [Art. 38(3)].
Result	Answer 1 **makes up a reason** (secrecy, Art. 38(5)). Answer 2 is correct: the DPO cannot be dismissed for doing their job.
Why	All the chunks about the data protection officer (Art. 37–39) look alike to the Bi-Encoder, so it ranked other DPO rules first. The answer also uses different words: the question says "fired", but the law says "shall *not* be dismissed or penalised", and the word "not" barely changes a vector. The Cross-Encoder linked "fired" to "dismissed" and ranked Art. 38(3) 1st.

Q4 – Buried at rank 4–10	
Question	Does a company have to stop processing if a person objects?
Answer 1 (Bi-Encoder)	The provided context does not directly state whether a company has to stop processing if a person objects. However, GDPR Article 6(1)(e) states that processing is necessary for the performance of a task carried out in the public interest or in the exercise of official authority vested in the controller. This implies that if an individual objects, the company might need to reconsider the necessity of processing their data for tasks related to public interest or official authority, but it does not explicitly mandate stopping processing. [Art. 6(1)(e)]
Answer 2 (Cross-Encoder)	Yes, if a person objects, the company has to stop processing the personal data unless there are compelling legitimate grounds for the processing that override the interests, rights and freedoms of the data subject [Art. 21(1-2)].
Result	Answer 1 **misses the rule** and cites the wrong article. Answer 2 is correct: stop, unless there are compelling legitimate grounds.
Why	The question uses everyday words ("company", "stop", "objects"), while the law uses legal words ("controller", "shall no longer process", "right to object"). The Bi-Encoder matched the common word "processing" and ranked general chunks about processing (Recitals 51 and 81, Art. 6(1)) first, leaving the right chunk 9th. The Cross-Encoder matched "stop processing" with "no longer process" and ranked Art. 21(1) 1st.

Q5 – Keyword trap (synonym)	
Question	How quickly must a company fix wrong information about me?
Answer 1 (Bi-Encoder)	The context does not directly address how quickly a company must fix wrong information about you. It discusses notifying authorities about personal data breaches but does not specify a timeframe for correcting inaccurate information. [Art. 33(1-2)]
Answer 2 (Cross-Encoder)	According to GDPR Article 16(1), a company must fix wrong information about you without undue delay. However, there is no specific timeframe mentioned for how quickly this must be done. [Art. 16(1)]
Result	Answer 1 **cannot answer**. Answer 2 is correct: wrong data must be corrected "without undue delay".
Why	"How quickly" matched the time words in the data-breach rules ("without undue delay", "72 hours"), so two of the Bi-Encoder's Top-3 were data-breach rules (Art. 33(1–2) and Recital 85). The real answer is the "right to rectification" (Art. 16), which never uses the words "fix" or "wrong", so it was only 9th. The Cross-Encoder understood that "fix wrong information" means "rectification of inaccurate data" and moved Art. 16 to 3rd, just inside the Top-3.

Summary: in all five questions the gold chunk was outside the Bi-Encoder's Top-3, so Answer 1 misstated a rule, missed an exception, made up a reason, or could not answer. The Cross-Encoder moved the gold chunk into the Top-3, and all five Answers 2 were correct.

Why does this happen? The Bi-Encoder turns each chunk into one vector *before* it sees the question, so it mostly matches the *topic* and the shared words. That causes three problems:

- **Same words, wrong meaning (Q1, Q5):** a chunk with the same word ("penalty", "how quickly") wins, even if it is about something else.
- **Different words, same meaning (Q3, Q4, Q5):** the right chunk uses legal words ("dismissed", "rectification") instead of the question's everyday words ("fired", "fix"), so it scores lower.
- **Small words get lost (Q2, Q3):** words like "not", "unless" or one key word like "encryption" hardly change a long chunk's vector, so the general rule and its exception look the same.

The Cross-Encoder reads the question and the chunk *together*, word by word, so it can see these connections. And because the LLM only reads the Top-3 chunks, a right chunk at rank 4–9 is never seen: the LLM then guesses or answers from the wrong rule.

3 Latency profiling

Measured on a laptop CPU (average of 10 runs per question for retrieval).

Step	**Pipeline 1**	**Pipeline 2**
Bi-Encoder retrieval (Top-10)	13.5 ms	13.5 ms
Cross-Encoder re-ranking (10 pairs)	–	292.8 ms
Total retrieval	**13.5 ms**	**306.3 ms**
LLM answer	50.5 s	40.3 s
End-to-end	**50.5 s**	**40.6 s**

Re-ranking makes retrieval about 23× slower, but it adds only 0.29 s, which is under 1% of the total time, because the LLM takes much longer.

4 Scaling question

The Cross-Encoder cannot pre-compute anything: it must read the question together with *every* chunk, again for every new question. At about 29 ms per chunk, all 559 chunks would take about 16 s per question (about 8 hours for 1 million chunks), while the Bi-Encoder searches all chunks in about 14 ms because their vectors are computed once in advance. So we use both: the fast Bi-Encoder picks 10 candidates, and the accurate Cross-Encoder re-ranks only those 10.